

BLENDER + UNITY / VERSION 1.0

Astral Wilds

Proof of concept & development blueprint



Avelora concept environment from the supplied presentation.

Make the first bond playable. A development plan for a fantasy creature RPG where a Magi explores Avelora, forms magical Contracts with Astrals, and uses a Conduit to shape tactical battles and companionship.

The deliverable to build: one repeatable Unity prototype that proves the Contract, Conduit and Bond loop, supported by a tested Blender asset pipeline.

Creative direction: Cameron Terrell McCarthy Jr.

Prepared: 15 September 2026

Intended readers: developer, designer, creature artist, animator, technical artist and producer.

Document status: proposed implementation and validation plan. The presentation, concept illustrations and opening film exist; this document does not claim a playable Unity game, finished creature rigs, test results or production funding. All gameplay values, asset budgets and schedules below are initial proposals for validation.

What this proof must establish

Turn a strong creative pitch into a small game that produces useful evidence.

The player promise

Become a Magi whose magic changes how Astrals fight, grow and open the world. The first memorable moment is a creature responding to the player; the next is a tactical choice that makes that relationship useful. The proof should connect those moments in one session.

Question	Evidence the prototype must produce
Does the encounter create curiosity?	Players notice a behavior cue, approach an Astral and understand an available action.
Does the Conduit add meaningful tactics?	Players can explain when protection, healing, a swap or a Contract was worth a scarce Focus charge.
Does a Contract feel readable?	New players finish the three-step ritual without spoken coaching and can identify why an attempt failed.
Does Bond motivate another expedition?	Players see a milestone reward, use it on a previously blocked route and select a next goal.
Can the team reproduce the visual direction?	One approved Blender creature survives export, Unity import, animation, VFX and a measured gameplay scene.

Three separate levels of evidence

Concept package: communicates the fantasy. **Proof of concept:** tests interaction and technical feasibility with readable placeholders. **Vertical slice:** demonstrates representative release quality for one short region. Approval of one level does not automatically approve the next.

Fixed direction and proposed decisions

The project name is Astral Wilds. Blender is the content creation tool and Unity is the game engine. The recommended first target is a premium-style, offline, single-player Windows prototype with keyboard/mouse and controller support. Platform expansion, price, release date and monetization are separate production decisions.

Success means a repeatable playable loop, a reproducible source handoff, and honest playtest findings - including reasons to revise the design.

How to use this blueprint

Read the gameplay contract first, then estimate implementation and asset production against it.

Pages	Read for
4-6	Scope, the first-session journey and the Valewyn test route.
7-13	Combat, starting values, abilities, Contracts, Conduits and growth.
14-19	Controls, state flow, software responsibilities, data, encounters and saving.
20-27	Unity setup; Blender modeling, rigging and export; presentation, performance and builds.
28-34	Milestones, prioritized work, functional tests, player research, risks, estimating and acceptance.
35-37	Canon alignment, unresolved decisions and primary technical references.

Decision labels

Established direction comes from the supplied game concept, deck or your instruction to use Blender and Unity. **Prototype proposal** is a concrete working choice introduced here so developers can build and test it. **Deferred** means deliberately excluded from this proof, even when it belongs to the full vision.

One integrated development loop



A developer should use this document to create a backlog and estimate, not treat its numbers as immutable balance. Changes that affect the player action budget, roster size, target hardware or completion criteria should be recorded in a short decision log before further content is made.

Review cadence

Demonstrate a runnable build every week once the first scene boots. Keep feedback in one ranked list. Each milestone review compares three things: agreed scope, observed player behavior and engineering or asset risks. A screenshot or edited film is useful supporting material but cannot substitute for the executable.

The page map refers to this PDF. Bookmarks provide direct navigation to each section.

The first-build contract

A small roster and one route are enough to test the connected systems.

Area	Proof-of-concept cap	Later vertical slice
Content	4 base species + 1 evolved-form proxy	6 base species + 2 evolved forms
Location	1 greybox route, camp interaction and reusable arena	Authored hub, wilds and ruin micro-region
Battles	4 active / 6 carried; 4 types; visible initiative	Tuned encounter set and 1 Order Trial boss
Magi	2 fixed loadouts; 3 Focus per encounter	Same loop with improved feedback and balance
Contracts	Readiness + 3 untimed choices; calm and weaken paths	Species-specific onboarding and presentation
Growth	1 Bond threshold, 1 gate interaction, 1 evolution proxy	Polished field reward and evolution presentation
Persistence	Local checkpoint save, party and storage	Full slice progression and recovery checks
Art	1 representative creature; proxies for the rest	Approved quality across the scoped section

What is actually counted

Four species can supply both friendly and enemy instances. A fifth shape used for evolution counts as an additional form, even if it reuses a rig. One camp is a menu and interaction area, not a housing simulator. The Order encounter is a test configuration of the same combat system, not a new boss framework.

Deferred from the proof

The 80-100 Astral ambition, ten-type balance matrix, broad branching evolution, Resonance transformations, Sovereign battles, mounts, online play, procedural world generation, crafting economy, farming, freeform building and a fully voiced story are outside this build. The UI may explain future growth without offering nonfunctional buttons.

Scope change rule: adding a species, body plan, traversal mechanic or combat subsystem requires an explicit estimate and a tradeoff against the current milestone.

The first 20 minutes

A proposed evaluation route, with a player decision at every major beat.

Beat	Player action and response	What to observe
0-2 min / Camp	Choose Warden or Mender. Meet Cindrel. Move, orbit the camera and inspect the Conduit.	Can the player identify their next destination?
2-5 / River path	Notice a Ripplefin cue. Inspect it, approach and enter a short wild encounter.	Does the cue make the Astral seem understandable?
5-8 / Contract	Calm or weaken a living target. Spend Focus and complete three attunement choices.	Is readiness clear? Does failure teach?
8-11 / Team test	Use a seeded four-Astral team in the arena. Spend scarce Focus on support or a swap.	Can the player follow front/rear and action order?
11-14 / Return	Visit camp, view Bond gained and claim Mossling's first field milestone.	Can the player separate Bond from combat strength?
14-18 / Ruin approach	Use the field reward at a root gate; replay a short Order test with a different loadout.	Does the reward create an obvious new goal?
18-20 / Anomaly	See an eleven-point sigil response. Save at camp and choose a next expedition.	Do they want another encounter, and why?

Timing is a test target, not a promise of content length. Include a clearly labeled developer test preset that supplies four party members and a reserve for the formation exercise. Do not disguise that seeded roster as progression the player has earned.

Two launch modes

Journey mode teaches the linked experience with a safe introductory encounter. **Systems test mode** starts directly in a fixed arena with known health, species, Focus and a reserve. Both call the same rules and data. The latter allows quick reproduction of tactical edge cases without replaying the story.

The intended emotional rhythm

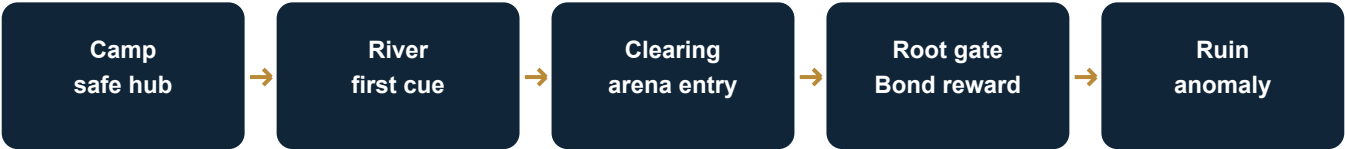
Curiosity at the river becomes tension in the ritual, relief when the creature joins, ownership in party selection and anticipation at the opened ruin. Keep the story glimpse short enough that the player remembers what their own actions accomplished.

Valewyn: one connected test route

An authored micro-region with a visible destination and a short return path.



Environment mood reference. The proof uses a compact greybox, not the pictured world scale.



Initial blackout envelope: approximately 120 by 80 Unity meters, with a 3-4 meter main path, a river edge that cannot be entered, one optional lookout and a shortcut back to camp. These dimensions are design hypotheses; adjust them using traversal time and camera tests before art production.

Space	Gameplay purpose	Required implementation
Camp	Loadout, healing, storage, Bond and save	One interaction point; clear safe-state UI
River edge	Read a creature before recruiting it	One patrol/sighting area and explicit encounter trigger
Clearing	Compare formation and Focus decisions	Reusable arena with 8 fixed creature anchors
Root gate	Make companionship change traversal	One persistent gate flag; Mossling ability prompt
Ruin	Deliver the eleventh-point hook	Short camera beat; optional skip; return objective

Layout checks before decoration

The next landmark must be visible from each decision point. Test the widest antlers at the arena camera, camera collision at the gate, ground contact on slopes and controller interaction at the camp. Use simple colliders and a reset-to-safe-ground fallback. Avoid making foliage obscure the only navigation clue.

A complete round of combat

The rules must make four active companions readable before animation becomes elaborate.

Step	Prototype rule
1. Round start	Clear expired effects; apply any scheduled start rules; award one action to each living deployed slot. Build and display the initiative queue.
2. Magi phase	Before Astral turns, optionally use one support, swap or Contract action. Every such action costs 1 Focus. Passing costs nothing and ends this phase.
3. Astral turns	Each living occupied slot acts once, fastest first. The player chooses an action and valid target at that slot's turn; enemies use their scored intent.
4. Resolution	Validate again, commit the action once, resolve numbers, then play presentation. Defeated or contracted targets leave the valid target set immediately.
5. Round end	Apply end-of-round rules, reduce cooldowns and test victory or defeat before starting another round.

Formation and initiative

Each side has Front Left, Front Right, Rear Left and Rear Right. While any living front Astral remains, basic melee cannot target either rear slot. Ranged actions may target any living enemy. Friendly support ignores formation. An empty front row exposes both rear slots. There is no automatic promotion from rear to front in this proof.

Sort initiative by Speed descending at round start. Break ties by alternating team priority each round, then fixed slot order. The queue belongs to the slots for the rest of that round. A swapped-in creature inherits the slot's queue position and remaining action; its own Speed takes effect next round. Do not silently reshuffle an already displayed queue.

End conditions

Victory occurs when no enemy remains active after defeat or a successful Contract. If no deployed ally remains alive, the player returns to camp, even if a reserve exists. This deliberately matches the pitch baseline and must be tested for frustration. Trainer/Order Astrals cannot be contracted. Retreat from a wild encounter returns to camp and grants no battle Bond.

All rules on pages 7-13 are proposed prototype specifications. Values are deliberately exposed for tuning.

Resolution that can be reproduced

Keep damage, status and targeting deterministic for early comparison.

Initial damage rule: $\text{damage} = \max(1, \text{floor}((\text{Power} + \text{Attack} - \text{Defense}) \times \text{type multiplier}))$. Barrier absorbs the result first; remaining damage reduces HP. HP stays between zero and maximum. A defeated Astral cannot act or receive healing. No critical hits, random misses, damage variance or same-type bonus in the proof.

Attack type \ target	Ember	Tide	Verdant	Terra
Ember	1.0	0.75	1.5	1.0
Tide	1.5	1.0	0.75	1.0
Verdant	0.75	1.5	1.0	1.0
Terra	1.0	1.0	1.0	1.0

Temporary four-type teaching matrix. Terra is deliberately neutral during the first balance pass. Only the three elemental techniques use these multipliers; a basic attack is neutral. Dual typing is disabled in the proof.

Worked example

Cindrel uses Ember Spark (Power 14, Attack 16) against Mossling (Defense 12). Base result is 18; Ember into Verdant gives $\text{floor}(18 \times 1.5) = 27$ damage. A 20-point Warden barrier absorbs 20, leaving 7 HP damage. The UI previews both values before confirmation.

Effect	Exact initial behavior
Guard	The next direct hit against the user before its next turn deals half damage, rounded down, minimum 1. Consumed by that hit. Applied before Barrier.
Barrier	Absorbs up to 20 damage. A new barrier replaces rather than stacks. Expires at the end of the current round.
Calm / observed	Encounter flags used for readiness only. No damage modifier, duration stack or ongoing crowd control.
Cooldown	A technique used in round R cannot be selected again in R+1. It returns in R+2. Basic and Guard are always available.

Invalid targets are rejected without spending the action. Once an action is committed, disable duplicate confirmation. If its original target is removed by a preceding effect, return to target selection where possible; never redirect a player attack to a random creature. Multi-hit attacks, simultaneous damage and damage-over-time are deferred.

A roster with four clear jobs

A proposed selection from the existing names, with one representative art target.

Species	Type / job	HP / ATK / DEF / SPD	Prototype role
Cindrel	Ember / pressure	80 / 16 / 8 / 14	Fast attacker; early companion; makes matchup advantage visible.
Mossling	Verdant / guard	100 / 10 / 12 / 8	Durable front-line option; carries the first field reward.
Ripplefin	Tide / ranged	85 / 13 / 9 / 12	Reaches protected rear targets; introductory wild Contract target.
Ironbur	Terra / anchor	115 / 12 / 15 / 6	Slow, neutral teaching opponent; tests defensive choices.

These values are a tuning seed, not validated balance. Set all units to one fixed combat tier. Do not add a level curve before the first battle study. Multiple instances of a species may be used in test encounters and reserves; each must still have its own individual identity.

Player-avatar allowance: one simple Magi proxy with locomotion and a casting pose, one Conduit prop and one follower at a time. Bespoke character creation, clothing simulation and a final-quality human rig are outside the proof.

The art production pilot

Choose **Mossling** as the first representative Blender asset because it tests quadruped locomotion, expressive ears, branch antlers, foliage and a useful interaction. Start from an approved interpretation of the named Mossling sheet. The separate realistic antler-stag image is a material and silhouette study, not automatically the same model.

One evolution form

Use a clearly labeled **Mossling evolved-form prototype**, with a changed silhouette and mesh reference but the same rig family. Its production name is unresolved. Keep base stats unchanged for this test so identity persistence and choice presentation can be tested independently of combat power. Other species show no evolution button.

Per-species design record

Record stable ID, display name, approved model sheet, scale, type, temperament cue, role, three equipped commands, readiness clue set, Bond reward, animation list, material family and estimated work. The artist, designer and developer approve the same record before integration.

The four-base-species cap is a systems requirement. It is not a commitment to four fully polished creatures before the gameplay has been evaluated.

The smallest useful ability kit

Teach distinct decisions without implementing the full future move library.

Command	User / target	Initial effect
Basic Strike	All / valid melee enemy	Neutral Power 10; no cooldown.
Guard	All / self	Halve the next direct hit until next turn; no cooldown.
Ember Spark	Cindrel / any enemy	Ember ranged Power 14; one intervening round cooldown.
Root Lash	Mossling / valid melee enemy	Verdant Power 12; one intervening round cooldown.
Tide Jet	Ripplefin / any enemy	Tide ranged Power 12; one intervening round cooldown.
Stone Ram	Ironbur / valid melee enemy	Terra Power 16; one intervening round cooldown.
Observe	Any allied Astral / any wild target	Reveal a behavior clue and set Observed once; uses that Astral's action.
Calm	Any allied Astral / any wild target	Set Calmed once if its revealed behavior response is selected correctly; uses that Astral's action.

Each Astral has the same two common combat commands and one species technique. Observe and Calm appear in a contextual wild-encounter menu rather than filling the normal technique bar. In Order tests those interactions are unavailable and explain why. At an individual unit turn the player performs exactly one command.

Data-driven action definition

Each ability record needs an ID, text key, category, target filter, Power, optional type, cooldown rule, effect identifier, animation cue and VFX cue. The resolver computes effects; an animation clip never decides whether damage happened. The UI obtains target legality and previews from the same rule service that executes the command.

Interface feedback

Before confirmation, highlight legal targets and show predicted HP damage, blocked damage and type multiplier. Disabled actions show a short reason: cooldown, no valid target or non-wild target. After execution, show one readable result label and update the queue. Include a battle log in the developer view.

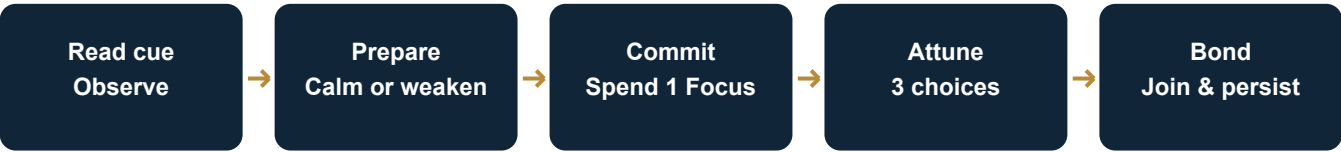
The source vision includes basic attacks, two techniques, a signature skill, passive traits and Resonance. This proof deliberately tests a reduced equipped kit. Expand only after players can explain the formation and Focus tradeoffs without interface overload.

Contracts: a readable ritual

Let observation and tactics prepare the same short act of attunement.

A target must be alive, wild, not already contracted, and eligible under its encounter rules. **Readiness starts at 0.** Observed contributes +1, Calmed contributes +1, and current HP at or below 50% contributes +1. Observed and Calmed are encounter flags and cannot be stacked. The HP contribution updates immediately when HP changes.

A Contract is available at readiness 2 or 3, with at least 1 Focus and an unused Magi action. Thus observation plus calming can avoid damage; observation plus weakening offers a battle path. Readiness enables the attempt rather than silently changing a random capture chance.



Ritual step	Player reads	Successful response
1. Element	Named type icon and text on the target	Select the matching sigil; prototype uses single typing.
2. Temperament	A visible behavior clue from Observe	Choose the taught response: offer space, approach gently or mirror a rhythm.
3. Alignment	A highlighted pattern in the Conduit ring	Select its matching symbol from three labeled options.

Combat pauses during the ritual. There is no timer or precision cursor test. One wrong choice ends the attempt with a specific explanation, consumes the spent Focus and returns to the next Astral turn. Correct choices do not consume additional Focus. Cancelling before the commitment confirmation is free; leaving the ritual after commitment counts as failure.

Initial cue mapping: Cindrel paws excitedly / mirror a rhythm; Mossling lowers its ears / approach gently; Ripplefin watches from the bank / offer space; Ironbur braces its stance / offer space. The same revealed response is used by Calm and ritual step 2.

Success and edge cases

Create a unique Astral instance once, mark the encounter spawn resolved, remove that target from combat, and add it to party if fewer than six are carried; otherwise use camp storage. It cannot enter the current battle. Full storage is outside the capped test, but the transaction must still fail safely if allocation fails. At 0 Focus, allow ordinary combat or retreat and explain that a later encounter refills Focus.

Tutorial rule: all required clues remain inspectable during attunement. Completion tests understanding, not memory or reflex speed.

The Conduit is the Magi's decision

Two fixed loadouts make prevention and recovery easy to compare.

Choice	Cost / timing	Benefit and limit
Warden: Ward	1 Focus, Magi phase	Give one living ally a 20-point barrier until round end. Does not heal.
Mender: Mend	1 Focus, Magi phase	Heal one living ally by 20 HP, capped at maximum. Cannot revive.
Swap	1 Focus, Magi phase	Exchange one living deployed ally with one living reserve. Inherit slot initiative and action state.
Contract	1 Focus, Magi phase	Attempt eligible wild recruitment; unavailable in Order encounters.
Pass	Free	Preserve Focus; ends the current Magi phase.

Choose the loadout at camp before the encounter. Each battle starts with exactly 3 Focus, with no in-battle regeneration and no Focus items. A round permits at most one Magi action, regardless of remaining Focus. The Magi deals no direct damage in this prototype.

Camp recovery is free and restores all owned Astrals to full HP. Between encounters, damage otherwise persists. Newly recruited Astrals retain their current HP until camp recovery; no consumable-healing inventory is included.

A decision worth testing

A front Astral has 12 HP and will receive 18 damage. Warden can prevent the knockout; Mender can restore HP first, but its value changes if a stronger hit is coming. A reserve Ripplefin can exploit a target but costs the same Focus as a Contract. Show enemy intent conservatively enough that protection can be an informed choice.

Avoid free-action loopholes

A reserve cannot be swapped into two slots, a fallen reserve is invalid, and swapping does not reset cooldowns or encounter flags on either individual. The outgoing Astral retains its HP and cooldown state. Cooldowns advance for carried reserves at round end as well as active units. Reserve effects that would alter combat are disabled.

Full-game direction

Core, Focus Crystal, Shaft, Rune Set and Catalyst remain the long-term Conduit components. Represent their flavor in one static inspection panel if useful. Do not build modular crafting, inventories for these parts or a full class tree until the two-loadout comparison demonstrates value.

Initial 20-point values are parity settings for testing, not evidence that healing and protection are equally strong in all encounters.

Bond, the field reward and evolution

Make progression visible, persistent and tied to a useful new possibility.

Bond is an individual relationship score from 0 to 100. It is distinct from HP and combat tier. For the prototype, grant +10 to each deployed surviving ally after the first victory in each unique encounter, +10 to a newly contracted Astral, and +10 for the first camp care interaction per individual. Do not grant repeatable Bond for reloading or farming the same completed encounter.

Milestone	Proposed implementation
Bond 20 / Mossling	Unlock Rootbridge, a contextual action at the one marked root gate. The gate remains open after use and is recorded in the save.
Bond 30 / Mossling	Offer the evolved-form proxy at camp after the ruin clue. Show a preview and confirm. Allow postponement without penalty.
Other prototype Astrals	Show Bond changes and a future milestone description, but no unimplemented active reward control.
Care interaction	One short animation and a relationship message. No hunger, neglect penalties or chore timers.

Field action contract

Rootbridge requires a living carried Mossling with the milestone unlocked, not necessarily the active follower. At the gate, show the required companion, preview the opened path and play a short contextual animation. Commit the gate flag before releasing player control. Re-entering the scene or loading a save must restore the open route without replaying rewards.

Evolution transaction

Preserve the individual ID, nickname, Bond, acquisition record and ownership. Change only form ID and presentation references in this proof. Set the evolution flag once and persist it. If the evolved prefab is missing, keep the current form and show an error in the developer log; do not create a second individual or lose the original.

The journey preset seeds Mossling at Bond 10 to reach one reward within the test session. Label that preset in the test report. A fresh-save progression test uses the ordinary award rules. The goal is to measure whether the reward creates a next objective, not to validate a long-term progression economy.

Controls and information design

Every required action must work without a mouse, sound cue or fast reaction.

Context	Keyboard / mouse proposal	Controller proposal
Explore	WASD move; mouse look; E interact; Tab journal	Left stick move; right stick look; south button interact; view/menu journal
Battle selection	Mouse select or arrows navigate; Enter confirm; Esc back	D-pad/stick navigate; south confirm; east back; shoulder buttons cycle details
Contract	Choose labeled options; Enter confirm; Esc opens cancel confirmation	D-pad choose; south confirm; east opens cancel confirmation
Pause / settings	Esc; clickable settings	Menu button; controller-navigable settings

Required screens

Start/continue and test-preset selection; exploration HUD with one objective; creature inspection; battle command and target panels; Focus and initiative display; Contract ritual; camp party/storage; Bond/evolution confirmation; pause/settings; save feedback. Each screen has an explicit default focus, back behavior and empty/error state.

Information hierarchy in combat

The current actor and next two queue entries are most prominent. Next show target validity, predicted result and remaining Focus. Formation slots remain stable. Creature type always includes a name and icon, not color alone. A target label must stay legible against both bright elemental effects and dark fur.

Accessibility baseline

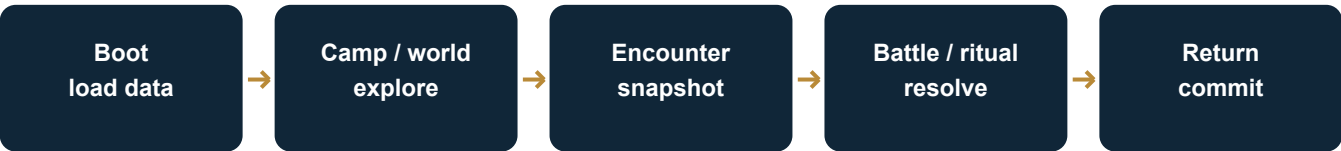
Provide subtitles for any spoken content, adjustable UI/text scale, independent music/SFX levels, camera sensitivity and invert-Y. Include reduced camera motion, optional damage shake and a skip control for presentation beats. Do not use a flashing ring as the only Contract cue. Pause freezes both world AI and combat presentation safely.

Input implementation rule

Use Unity Input System action maps for Explore, Battle, Contract and UI, plus a small always-available pause action. Enable only the active context and route UI through one event system. Clear the previous selection when switching contexts so one confirm press cannot both open a menu and activate its first item. Action-based input is supported by Unity's Input System documentation [S7].

Scenes and game states

Use explicit transitions so exploration, battle and saving cannot disagree.



State	Owns	Exit requirement
Boot	Version check, data validation, local save load	Valid definitions and recoverable save state
Explore	Movement, camera, follower, encounter eligibility	Freeze controls and capture world return location
Battle setup	Party snapshot, slots, encounter seed	All required actors and UI ready
Magi / Astral turn	Legal commands, queue and committed results	Exactly one resolution or explicit pass
Contract	Paused battle, attunement UI and outcome	Success/failure transaction completed once
Results / return	Rewards, resolved spawn IDs, checkpoint	Write safe state and restore world input
Camp menu	Storage, loadout, care and evolution	Validation complete before close/save

Scene plan

Use Boot for persistent services; Valewyn_POC for the camp and route; Battle_POC for the reusable arena; and AssetLab for isolated model, lighting and animation tests. Keep only one gameplay camera/audio listener active. Load the arena through a short controlled transition; seamless battles are not required to prove the design.

Ownership at transitions

A session coordinator owns the current game state and transition lock. Battle receives a snapshot of the selected party, encounter and return point, then returns an explicit result. It does not modify random objects left in the exploration scene. A second trigger during setup is ignored. On load failure, return to the previous safe state with input restored.

Failure behavior

If a visual effect or animation is missing, show a simple fallback and finish the already resolved action. If required gameplay data is invalid, stop entry to that encounter and present a recoverable error. Never leave the player in an endless transition because a cosmetic event failed.

Unity system responsibilities

Keep game rules independent from the objects that display them.

Module	Responsibility	Must not own
BattleDomain	Pure C# state, commands, initiative, target rules and damage	Scene loading, animation or UI widgets
BattlePresenter	Spawn views and display already resolved outcomes	Authoritative HP, chance or reward decisions
ContractService	Eligibility, clue checks and recruitment transaction	Arbitrary direct mutation of party UI
RosterService	Unique individuals, party slots, storage and evolution	Per-frame rendering or navigation
ProgressionService	Bond awards, reward IDs, gate and story flags	Re-awarding completed objectives on scene load
SaveService	Versioned snapshots, validation and recovery	Serializing live scene object references
SessionCoordinator	State changes, locks, setup and safe return	Duplicated battle calculations
Input / UI adapters	Map commands to domain requests; show previews	A second implementation of target legality

A command travels in one direction

Input requests `UseAbility(actorID, abilityID, targetID)`. The domain validates phase, cost, cooldown and target. It produces an ordered result containing damage, state changes and presentation cues. The presenter plays that result, acknowledges completion and unlocks the next decision. An animation timeout can release presentation without repeating domain effects.

Definitions and instances

Use `ScriptableObjects` for authored species, abilities, encounters and loadouts. Create independent runtime data for each individual and battle. Unity documents `ScriptableObjects` as shared asset data; they are not the save system for changes made by a deployed game [S4]. In particular, damaging one Mossling must not modify the shared species definition or every other Mossling.

Practical implementation boundary

Use ordinary `GameObjects`, `MonoBehaviours` and a small service composition at Boot. Split domain, presentation and tests into clear assemblies. Avoid introducing networking, an entity-component rewrite, a general quest language or a complex dependency framework unless a measured requirement demands it.

The data the prototype needs

Stable identifiers make repeatable tests and future content expansion possible.

Record	Required fields
SpeciesDefinition	speciesId, displayNameKey, typeId, baseStats, abilityIds, clueSetId, prefabId, bondRewardId, allowedFormIds
AstralInstance	individualId, speciesId, formId, nickname, currentHP, bond, acquiredEncounterId, earnedRewardIds
PartyState	ordered carriedIndividualIds (max 6), four deployedSlotIds, storageIndividualIds, loadoutId
BattleState	battleId, round, phase, queueSlotIds, actionConsumedBySlot, Focus, active unit states, cooldowns, readiness flags
EncounterDefinition	encounterId, spawnId, enemy species/forms and slots, AI profile, contractAllowed, firstWinRewardId
WorldProgress	completedEncounterIds, openedGateIds, storyFlags, lastSafeLocationId
SaveEnvelope	schemaVersion, buildId, contentVersion, timestamp, profileId, roster, party and world progress

ID and validation policy

Use durable string IDs such as species.mossling and gate.valewyn.rootbridge. Give individual Astrals generated unique IDs. Do not use an asset name, list index or Unity scene instance ID as persistent identity. Definitions may change names without changing their IDs. Duplicate IDs, missing references and invalid slot counts fail the content validation check before a build is handed over.

Example of an identity-preserving operation

Individual aw-0017 starts as species.mossling, form.base, Bond 30. Evolution updates formId to form.mossling.poc_evolved and records the evolution reward. Its individualId stays aw-0017. The party still references aw-0017, so storage, Bond and quest references remain intact.

State that should stay transient

Current cursor selection, animation state, particle instances and camera interpolation do not belong in the save. Battle state is replayable in developer tests but is not a resumable user checkpoint in this proof. Save only at the stable boundaries defined on page 19.

Encounter design and enemy behavior

Create controlled comparisons before adding an extensive creature ecology.

Encounter	Initial composition	Question tested
River introduction	1 wild Ripplefin; low pressure; recruitable	Can a first-time player read, prepare and Contract?
Formation lab	4 allies vs 2 front Ironbur + 2 rear Cindrel	Can players identify blocked melee and use ranged reach?
Focus dilemma	Seeded wounded front ally, living reserve, 1 Focus; wild target at readiness 2	Can players explain support versus swap versus recruitment?
Order rehearsal	4 vs 4 using the same species pool; recruitment disabled	Can the loop work as an authored tactical challenge?

Deterministic battle AI

Score legal actions at an enemy's turn. Prefer an available technique if it deals at least 5 more predicted damage than Basic; otherwise prefer Basic. Choose the valid target with the lowest post-hit HP, using fixed slot order for ties. If below 25% HP and Guard was not used on its prior turn, use Guard. This is a debugging policy, not an intended final difficulty model.

Show a simple intent category before the player commits: attack, heavy technique or guard. If a target is removed, re-evaluate legality at the enemy turn. Log the chosen action, candidate scores and reason in the developer overlay. No enemy Magi system is required for the first lab; the Order trainer is presentation around the shared encounter rules.

Exploration behavior

Use authored patrol points and four visible states: Idle, Notice, Cautious and EncounterReady. Distance and line of sight trigger transitions with a short debounce so states do not flicker. Avoid a full feeding, breeding or migration simulation. Navigation can begin with simple waypoints; add the compatible Unity AI Navigation package only if obstacle avoidance in the route requires it.

Tuning with control

Keep the same initial team, positions and encounter state when comparing Warden with Mender. Record build and content versions. Change one major rule at a time, then rerun the same fixture before drawing conclusions from a new test group.

Saving without losing a companion

Recruitment and evolution must survive an interrupted session.

Use one local profile plus a previous-good backup. Save at camp, after encounter results, after a field gate opens and after evolution confirmation. Disable manual saving during combat and the ritual with a clear message. This does not mean progress is disposable: a successfully completed Contract must be recorded as a recruitment checkpoint before play resumes.

Recruitment checkpoint

Commit the individual, acquisition event and resolved spawn ID together in a stable snapshot. If the process closes before the rest of that battle ends, reload at camp with the contracted Astral retained and the resolved wild spawn absent. Do not attempt to resume the partially finished battle. This makes the recovery policy explicit and prevents duplicate recruitment.

Write and recovery sequence

Build a serializable snapshot; validate unique identities, party membership, known definition IDs and value ranges; write a temporary file; flush and close it; read it back; replace the active save through a tested same-volume safe replacement while retaining the previous good version. On startup try the active snapshot, then the backup. Explain any recovery instead of silently creating a fresh game.

Case	Required result
Same Contract completion signaled twice	One individual and one reward event only.
Evolution followed immediately by reload	Same individual ID, new form, unchanged Bond and party location.
Missing species in a newer content build	Stop destructive overwrite; show version error and retain recoverable file.
Invalid or truncated active file	Load previous-good backup if valid; report the recovery.
Party full at recruitment	New individual goes to storage; no forced deletion or duplication.
Checkpoint write fails	Keep in-memory result; report failure and allow retry; do not claim saved.

Use explicit schemaVersion migrations once the schema changes. For this proof, refusing an unknown newer schema is safer than guessing. Store files through the platform-appropriate persistent data location. Cloud sync, account services and encryption for anti-cheat are deferred.

The Blender + Unity baseline

Pin a working toolchain after a small compatibility test.

Tool / decision	Proposed project policy
Unity editor	Use a supported Unity 6 release. Default for a new locked proof: Unity 6.3 LTS, after package and import checks. Unity's current support page lists support through December 2027 [S1].
Installed Unity	6000.6.0f1 was found on this machine. That is installation evidence, not proof of an existing project or validated package stack. If a project already uses it, test that project in place before choosing a different baseline.
Blender	Blender 5.2.1 LTS was used locally for the concept-film encoding. Pin the chosen Blender build for asset production only after the mesh/rig round-trip test.
Rendering	Use Universal Render Pipeline (URP) for the first PC prototype. URP is Unity's configurable render pipeline [S5]; the choice here prioritizes a controlled stylized rendering setup.
Code and source	C# gameplay; Git plus large-file storage for binary assets; explicit editor and package versions; Windows executable for first tests.

Initial project setup

Create the URP project only after checking whether there is an existing project to preserve. Configure product name Astral Wilds, organization identifier supplied by the owner, linear color workflow, input actions, explicit quality settings and the four test scenes. Lock package manifest and package lock alongside project settings. Record every selected package version in the build README.

Packages chosen for a purpose

Input System supports the two input schemes; the Unity Test Framework supports rules and scene tests. A compatible Cinemachine release may simplify camera work. Timeline is optional for the short ruin beat. Use built-in particle systems first. No Addressables, online services or third-party capture system is required for the tiny local proof.

Do not open a newer Unity project in an older editor as a shortcut. Version changes need a backup/branch, import test and runnable-build check. No engine installation or project migration is performed by this PDF.

Blender: build one convincing Astral

Approve the silhouette and deformation before investing in surface detail.



Named Cindrel and Mossling sheet. Resolve proportions and life stage before modeling.

Stage	Artist deliverable	Review question
1. Design lock	Front/side/back, scale, palette, mouth/eye plan and neutral pose	Is this the same creature from every angle?
2. Blockout	Low-detail body, ears, antlers and major foliage masses	Does it read at the battle camera distance?
3. Game mesh	Clean deformation topology, UVs and material slots	Can legs, jaw and ears bend without collapsing?
4. Surface pass	Base color, normal detail, masks and emission where needed	Does the material identity survive reduced detail?
5. Integration	Exported mesh, rig, clips and Unity prefab pilot	Does the creature work in the actual scene?

Modeling rules for the pilot

Author at real scale: one Blender meter should import as one Unity meter. Keep a human-height reference beside the creature and record shoulder height. Put the intended root at ground center. Resolve rotation and scale before rigging; avoid negative scale and last-minute transform changes after animation starts.

Treat the artwork's fur, leaves and glowing details as a style target, not a requirement for individual strands or leaves. Use larger readable clumps, baked detail and a few controlled cards. Keep silhouette-defining antlers as geometry; do not build hidden detail that the gameplay camera cannot reveal.

Texture and material handoff

Deliver clearly named texture files and a material specification. Bake procedural details into portable maps where needed. Blender node graphs are not an automatic Unity shader conversion. Rebuild the approved look in URP using a small shared material family; compare it under the AssetLab light rig [S2, S3].

Blender: rig and animation plan

A compact clip set should express personality and satisfy the game's state changes.

Use a deform skeleton separate from animator controls. For a quadruped, include root, body/spine, neck/head, jaw, leg chains, tail and ears; foliage and antlers need only the movement their design justifies. Bake controller/constraint results to export bones. Constraint logic itself should not be expected to run in Unity [S2].

Clip	Initial duration / mode	Acceptance focus
Idle	3-5 sec, loop	Breathing and ears; stable feet; no root drift
Walk / Run	1-1.5 / 0.6-1 sec, loop	In-place speed matches controller; clean loop seam
Notice / Calm	0.8-1.5 sec each	Readable emotional shift at game distance
Basic / Technique	0.7-1.3 sec each	Anticipation, impact and recovery; clear target direction
Hit / Guard	0.3-0.6 / 0.8 sec	No ground penetration or unreadable camera movement
Defeat	1-1.8 sec, hold end	Safe final pose; collider/presentation agreement
Contract / Celebrate	1-1.8 sec each	Distinct joining moment and stable transition to idle
Rootbridge / Care	1-2 sec each, pilot only	Align to interaction marker; safe interruption

Animation scope arithmetic

The shared minimum is 12 clips per rig: Idle, Walk, Run, Notice, Calm, Basic, Technique, Hit, Guard, Defeat, Contract and Celebrate. The Mossling pilot adds Rootbridge and Care. Four unique rigs would imply up to 50 clip assignments before the evolved-form review. Reuse where anatomy genuinely permits; do not assume a biped, quadruped and aquatic body share the same motion.

Import and playback contract

Use Unity Generic rigs for non-humanoid Astrals, with an explicit root node [S6]. Prefer in-place locomotion in the proof, with movement controlled by game code. Record clip names, ranges, loop flags and root-motion settings in a manifest. Visual animation events may trigger footsteps and effects, but authoritative damage still comes from the battle resolver.

Test idle-to-run, attack-to-hit, defeat during an effect, and scene unload during celebration. Every animation must have a fallback path to release input and finish presentation.

The asset handoff: .blend to prefab

Make export repeatable rather than repairing each import manually.



Checkpoint	Starting convention and test
Source preparation	Keep editable .blend source outside Unity Assets. Export selected mesh and deform armature only. Use named collections and an explicit export manifest.
FBX preset	Start with scale 1, -Z Forward and Y Up; export mesh/armature, disable leaf bones and bake required clips. Confirm actual size and facing with a one-meter test cube and an asymmetric animated test mesh.
Transform review	Do not compensate for an incorrectly scaled rig by scaling every Unity prefab. Correct the source/export preset. Confirm ground root, facing, bind pose and deformation before locking the preset.
Unity model import	Review scale, normals/tangents, Generic rig root, clips and material remaps. Use a repeatable import preset and explicit texture references [S3, S6].
Prefab assembly	Add Animator, visual root, simple collider, view component, species reference, VFX attachment points and audio anchors. Keep game state outside the model.
Round-trip check	Change one clip or texture in Blender, re-export, and confirm the prefab retains gameplay references and appearance.

Why exported interchange files

Unity supports FBX as its main model interchange workflow. Native .blend import depends on Blender being installed on the importing machine; exported FBX avoids that dependency in the shared Unity project [S3]. Store source and exported assets so another developer can reproduce both the build and the art handoff.

Pilot acceptance checklist

Correct meter scale and forward direction; feet at ground; no extra leaf bones; no missing maps or pink materials; all required clips present; no bound-pose distortion; eyes readable; shadows acceptable; prefab runs in a standalone build; source export can be repeated by a second person using the written preset.

Axis and bake settings above are a starting preset, not a universal fix for every existing rig. Validate them against the pinned Blender/Unity versions before batch export.

Environment, lighting and readability

Build the test route from a small reusable kit.

Asset family	Initial proof allowance	Production strategy
Ground and paths	1 terrain/ground layout; 2-3 surface looks	Author readable slopes and paths first; avoid a complex terrain toolchain.
Trees and foliage	2 tree silhouettes; 3 shrubs/ground clusters	Reuse meshes/materials, vary scale within limits; keep the interaction path clear.
Rock kit	3 rock forms; 1 ledge module	Simple collision proxies; repeatable meter dimensions.
Ruin kit	Wall, pillar, arch, step and sigil plinth	Grid-compatible modules; one focal anomaly location.
Camp and gate	1 Conduit station; 1 care point; 1 root gate	Interactive props get clear anchors and state variants.

Lighting baseline

Use one key directional light and a restrained environment fill. Start with a fixed time of day. Author local magical glow with emission and a limited number of purposeful lights; a glowing texture does not need its own real-time light. Evaluate creatures in shade as well as direct light. Keep UI and target indicators readable against both.

Materials and texture intent

Use a consistent roughness/smoothness interpretation across the material library and document any packed channels. Mark base color/emission and non-color data maps appropriately at import. Match tangent-space normal orientation through the pilot. Avoid building the visual target on a shader graph that only one creature can use.

Camera and collision checks

The exploration camera must not pass through the root gate or bury itself in leaves. Arena composition must accommodate a large Ironbur and Mossling antlers without cutting off target indicators. Test the screen at 1280 x 720 as well as 1920 x 1080; the creature should remain identifiable without opening a close-up inspection view.

Approve the environment in gameplay lighting and camera positions. A Blender beauty render alone does not establish the Unity visual target.

Magic, sound and cinematic moments

Presentation should make rules easier to feel and understand.

Moment	Visual direction	Audio / interaction rule
Contract ready	Stable sigil and explicit readiness label	Soft cue; no dependence on hearing it
Contract commit	Three readable ring stages with one highlight	Distinct step feedback; untimed; skip decorative movement
Barrier / heal	Different silhouettes: shell versus rising motes	Separate short sounds; show prevented or restored value
Technique hit	Element-colored effect with clear impact point	Short transient; avoid masking subsequent actor cues
Bond reward	Companion reaction and concise milestone card	Gentle motif; confirmation opens next objective
Eleventh point	Extra sigil point and a brief ruin response	Low restrained swell; skip returns to a safe playable state

A small effects library

Build one reusable projectile, one contact burst, one barrier shell, one heal effect, one ring segment, one Bond accent and one root-growth placeholder. Expose color, duration and attachment point as data. Pool frequently spawned effects if profiling shows repeated allocation costs. Avoid simultaneous full-screen flashes and heavy transparent layers over four active creatures.

Audio production minimum

Cover footsteps, UI confirm/back/error, Contract stages, basic impact, four technique identities, barrier, heal, Bond reward and one ambient bed. Add creature vocalizations only for the representative pilot initially. Route Master, Music, SFX and UI separately. Record source and permitted use for every audio file, including any commissioned or synthesized work.

How the existing film is used

The opening MP4 is a pitch and mood reference made from concept art with camera movement and music. It is not a gameplay capture or a substitute for animated game-ready creatures. The Unity proof can omit it from the default boot flow. If included as an optional menu extra, make it skippable and return reliably to the menu.

Use a short in-engine ruin beat for the proof, with a reduced-motion option. Reserve elaborate Sovereign cinematics and Resonance sequences for separately estimated later work.

Performance targets and measurement

Choose a reference machine before promising frame rate or asset density.

Proposed reference class: Windows 11, six-core desktop CPU, 16 GB RAM, discrete GPU with 6 GB VRAM and SSD. This is a planning class, not a verified minimum specification. The team must name the actual CPU, GPU, driver, RAM and display resolution in the first performance report.

Area	Initial target / starting cap	How it is evaluated
Frame pacing	1080p; 60 fps goal; 95th-percentile frame time at or below 16.7 ms in the fixed test	Capture a 3-minute route plus worst-case 4v4 scene; report loading separately.
Creature geometry	15k-30k triangles for representative LOD0; simpler proxies elsewhere	Measure full scene; reduce hidden detail before losing silhouette.
Creature rendering	1-2 materials; one 2K texture set as a starting budget	Measure draw calls, texture residency and transparency cost.
Rig complexity	About 40-70 deform bones for the pilot; target max 4 weights/vertex	Validate deformation and skinned-mesh cost with eight creatures visible.
Memory	Initial investigation threshold: 3 GB process memory and 4 GB VRAM	Record actual peaks and platform reporting method; these are not measured results.
Loading / stability	Warm scene change under 5 sec; 30-minute repeat test without a blocker	Record machine, build mode, repeated loop and any stalls.

Measurement procedure

Create a reproducible performance preset with the same camera, actors, lights and effects. Use a Development Build and Unity Profiler for diagnosis, then repeat timing in the intended release build to account for instrumentation overhead. Record CPU and GPU frame times separately rather than adding them as if they execute serially [S8].

Optimize the measured bottleneck: expensive shadows, overdraw, excessive materials, animation updates or allocation spikes. Add LODs and culling when captures justify them. Keep comparison screenshots and profiles so an optimization does not silently damage readability.

Every number on this page is a proposed budget or threshold. The proof has no established benchmark results yet.

Source control and reproducible builds

Another developer should be able to continue without rebuilding the project from memory.

Location	Purpose
UnityProject/Assets/_AstralWilds/	Scenes, Scripts, Data, Prefabs, Art, Audio, UI and Tests.
UnityProject/Packages + ProjectSettings	Committed dependency manifest, package lock and settings.
ArtSource/Blender/	Editable .blend files, export presets and rig/clip manifests.
ArtSource/Textures/	Editable texture sources and exported maps, with naming conventions.
Docs/	Setup guide, decision log, asset register, test protocol and milestone records.
Builds/ and TestReports/	Versioned deliverables outside the ordinary Unity asset tree.

Version-control rules

Commit Unity .meta files with their assets so GUID references survive transfer. Use visible metadata and text serialization for supported Unity assets. Exclude generated Library, Temp, Logs and local build caches. Track large binary sources such as .blend, high-resolution textures and video through the chosen large-file policy. An exported FBX does not replace the editable Blender source.

Build identification

Each test build displays version, commit ID and content version in the pause menu. Archive its exact rule data, test presets and known issues. The handoff records Unity editor version, Blender version, package versions, platform/backend, build scene order and how to create a fresh test save.

Release procedure for each milestone

Validate content IDs, run domain tests, run the scene smoke test, build Windows, launch outside the Editor, play from a clean save, then test loading the supplied fixture save. A second person checks the written setup from a fresh checkout. Keep a last-known-good build for comparisons; a new film or screenshot is never used as evidence that a broken build is acceptable.

Review discipline

Use short branches and small reviewable changes. Commit rule changes separately from large art drops where practical. Every approved scope change has an owner, reason, affected tests and estimate impact. Asset renames and folder moves occur inside Unity where possible, with references checked afterward.

Milestones and a plausible sequence

An illustrative schedule to estimate with a team, not a delivery commitment.

Gate / window	Work and dependency	Exit evidence
G0 / weeks 1-2	Lock scope, reference hardware, versions and roster. Run a primitive mesh/rig round-trip immediately.	Decision log, priced backlog, valid Unity boot build, tested export preset.
G1 / weeks 3-6	Build deterministic battle rules, targeting, Focus and the four-species data set. Start Mossling art pilot in parallel.	Repeatable arena; domain tests; four active slots and reserve behavior demonstrated.
G2 / weeks 7-10	Connect route, Contract, party/storage, Bond, gate, evolution proxy and checkpoints.	End-to-end journey with both input schemes; save recovery fixtures pass.
G3 / weeks 11-14	Integrate representative art, animation and feedback. Conduct pilot usability sessions, then unfamiliar-player study.	AssetLab approval, performance captures, raw observations and defect triage.
Revision allowance / weeks 15-20	Bounded changes driven by failed tests, asset rework or toolchain risk. Re-test changed areas.	Accepted prototype or explicit revise/stop decision; separately priced slice plan.

This sequence assumes roughly four full-time-equivalent contributors with the required skills, availability for reviews and a restrained art target. Some roles may be part-time specialists. The schedule must be replaced by the partner's estimate after G0. A solo developer will need a smaller scope or substantially longer calendar time.

Critical path

Target rules and action ownership precede balanced encounters. Stable individual IDs precede recruitment and evolution persistence. The export pilot precedes a roster art batch. An end-to-end build precedes unfamiliar-player testing. Do not wait until week 11 to discover that the rig or Unity import workflow is unusable.

Milestone review outputs

For every gate: executable, source revision, short demonstration, acceptance checklist, unresolved defects and a cost-to-complete update. The creator consolidates feedback within an agreed review window. A missed gate prompts a specific correction or scope reduction rather than an automatic extension into more content.

28 / PRIORITIZED WORK

A backlog a developer can estimate

Start with observable outcomes and dependencies.

ID / priority	Work item and completion condition	Depends on
P01 / Must	Boot + test preset: standalone build opens the route and arena without missing references.	Toolchain lock
P02 / Must	Species/instance data: four definitions, unique individuals and validation errors for duplicates.	P01
P03 / Must	Battle resolver: initiative, targeting, damage, Guard and cooldown fixtures pass.	P02
P04 / Must	Magi actions: 3 Focus, one action per round, Warden/Mender and legal swaps work.	P03
P05 / Must	Contracts: both readiness paths, three choices, failure and one-time recruitment work.	P02-P04
P06 / Must	Route + state flow: encounter, return, retreat and camp input transitions work.	P01, P03
P07 / Must	Bond/gate/evolution: one reward path preserves identity and exposes a next goal.	P05-P06
P08 / Must	Checkpoint persistence: recruit, evolve, gate and backup recovery fixtures pass.	P02, P05-P07
P09 / Must	Input/UI: all required actions complete on controller and keyboard/mouse.	P03-P08
P10 / Must	Blender pilot: Mossling model/rig/clips integrate and re-export cleanly.	Early export test
P11 / Should	Representative audio/VFX and short skippable anomaly beat.	P06, P10
P12 / Must	Validation package: player sessions, performance report, known issues and handoff.	P01-P10

What to cut first if the estimate grows

Reduce environmental decoration, clip variety on proxy creatures and optional cinematics before cutting the Contract, Focus decision or identity-preserving save path. A fourth body plan can stay a proxy while still testing a fourth combat role. A reduced prototype must still run the promised core tests.

Estimate each ticket

Record owner, effort range, review time, external dependency, acceptance test and risk. Split a ticket if it cannot be demonstrated within a normal weekly review. Mark prototype shortcuts explicitly so they are not mistaken for approved production architecture.

Functional acceptance tests

A playable prototype must pass these before audience feedback can be interpreted.

ID	Test setup / action	Pass condition
F01	Load four active plus one reserve; play two rounds.	Every living slot acts once per round; queue is stable.
F02	Melee targets rear with a living front; repeat with no front.	First is rejected free; second is legal.
F03	Execute worked damage example on page 8.	27 damage; 20 absorbed; 7 HP lost.
F04	Use three Magi actions over rounds, then request a fourth.	Focus reaches 0 and action is disabled with reason.
F05	Swap a living reserve into a seeded consumed slot.	No additional turn; cooldown and individual HP preserved.
F06	Observe+Calm, then Observe+weaken on separate targets.	Both reach readiness 2 and permit a Contract.
F07	Choose one wrong ritual answer; retry next round.	One Focus spent; specific feedback; battle resumes.
F08	Finish Contract while party is full; duplicate completion event.	One stored individual; one resolved spawn ID.
F09	Close after recruitment checkpoint; reload.	Camp resumes with retained companion and no duplicate spawn.
F10	Care, win unique encounter, open gate, evolve, reload.	Rewards awarded once; gate open; ID and Bond retained.
F11	Corrupt active test save; keep a valid backup.	Backup recovery is reported; valid data preserved.
F12	Complete journey on controller, then keyboard/mouse.	No blocked screen, missing focus or double confirmation.
F13	Skip ruin beat; pause during effects; defeat all active allies.	Safe control restoration and expected camp return.
F14	Re-export Mossling and build from a fresh checkout.	Clips/materials/references intact; standalone build runs.

Use automated domain tests for arithmetic, targeting, action ownership and transactions; Play Mode tests for transitions and references; manual checks for visual clarity, controller navigation and sound. The prototype cannot pass by having only a successful Editor play session.

All entries are planned tests. No pass results are asserted in this PDF.

Test with people who have not seen it

Small-sample observation is a design gate, not proof of market demand.

Run two internal pilot sessions to remove broken instructions, then ten unfamiliar-player sessions on one frozen build. Recruit people comfortable with adventure or RPG controls, while including several with little creature-RPG experience. Record familiarity and input device. Exclude the pilot sessions from the ten-player counts.

Measure	Proposed decision target	Evidence
First Contract	At least 8 of 10 complete unaided	Success, elapsed time, wrong steps, requests for help.
Conduit understanding	At least 7 of 10 explain a concrete protection/healing tradeoff	Verbatim explanation after play, assessed against a short rubric.
Next goal	At least 7 of 10 select the opened route or another valid next goal without prompting	Chosen objective and the clue they cite.
Battle readability	Investigate any repeated inability to identify current actor or legal target	Confusion events, selection reversals and decision time.
Motivation	Exploratory: ask whether they would play another encounter and why	Reasons and observed choice; no sales or retention claim.

Session protocol

Use the same objective: form a Contract, try the team battle and discover what the new bond makes possible. Observe the first attempt without verbal coaching. If a blocker forces assistance, record where and why; do not count assisted completion as unaided. After play, ask what Focus did, why a target was blocked, how Bond differed from health, and what they would do next.

Instrumentation

Record pseudonymous participant ID, build/content version, device, encounter start/end, action choice, target rejection, Focus use, readiness change, Contract step/result, Bond reward, gate use, save result and crash/blocker. Log elapsed times without personal account data. Explain recording to participants and keep only information needed for the study.

Interpret the result honestly

Report counts and individual observations, not only percentages. Ten people cannot establish market size or long-term retention. When a target fails, classify whether the cause is rules, communication, technical failure or player preference. Price one bounded revision and retest with new players before broadening scope.

The risks that should shape spending

Give each major uncertainty a test, an owner and a fallback.

Risk / owner	Early warning	Response and fallback
4v4 pacing / Design	Players lose track of the actor or wait through long effects.	Shorten presentation, simplify kits and clarify queue. Evaluate a 3v3 comparison only as an explicit change to the pitch baseline.
Contract frustration / Design	Players burn Focus without understanding readiness or the failed choice.	Keep clues visible, improve feedback and reduce ambiguity before adding timing pressure.
Focus exhaustion / Design	Players cannot recruit after necessary defensive actions.	Test encounter pressure and reset policy; compare a separately versioned cost change rather than secretly gifting Focus.
Art cost / Art lead	Pilot rig, antlers or foliage exceed the intended effort or GPU budget.	Simplify material detail and secondary motion; keep other species as proxies until the pilot is repeatable.
Pipeline mismatch / Technical art	Scale, materials or clips require repeated manual repair.	Lock a tested export/import preset; reject the batch until one full round trip passes.
Persistence defects / Engineering	Duplicate recruits, lost Bond or changed IDs after evolution.	Stop new progression work; fix transaction tests and recovery before player sessions.
Scope growth / Producer	More species or systems are requested before a failed gate is resolved.	Trade against approved work or price a later milestone; update acceptance criteria.
Visual identity / Creative lead	References disagree on starter stage, anatomy or type.	Approve a per-species model sheet and canon decision before final production.

Stop or revise signals

Do not proceed to a polished slice while recruitment duplicates data, the build cannot be reproduced, the main interaction remains incomprehensible after a bounded revision, or representative asset cost is still unknown. These conditions require a concrete decision, not additional concept art.

The project owner prioritizes creative goals; the developer owns technical estimates and risk reporting. Both approve any change that alters what the proof claims to demonstrate.

People, effort and estimating

Price the scope from work packages rather than the size of the world vision.

Capability	Illustrative effort range	Coverage
Gameplay / tools engineering	18-24 person-weeks	Rules, input, UI integration, saves, builds and automated checks.
Systems / level design	8-12 person-weeks	Rules, encounters, route, onboarding, tuning and study interpretation.
Creature / technical / environment art	13-18 person-weeks	One representative creature, proxies, export pilot, rig/animation, kit and presentation.
QA / production / audio support	8-12 person-weeks	Build checks, coordination, test sessions, audio implementation and delivery.
Base effort total	47-66 person-weeks	Not a staffing commitment or supplier quote.
With 20% planning allowance	About 56-80 person-weeks	At four FTE, about 14-20 weeks before specific scheduling constraints.

A person-week means approximately one person's full working week. The ranges assume existing professional skills, limited original environment work and proxies for most creatures. They are rough planning assumptions to challenge during discovery. Part-time specialists, procurement, review delays and unfamiliarity with rigging can lengthen calendar time.

Cost worksheet for the development partner

For each work package: estimate low/high person-weeks, multiply by the agreed fully loaded weekly rate, add explicitly quoted external art/audio/testing and tool costs, then show contingency separately. State the quote currency, tax treatment, payment milestones and assumptions. This document deliberately does not invent a funded budget or a fixed release date.

Measure the first creature's actual cost

Track design cleanup, model, UV, texture, rig, animation, export repair, integration and revision time separately. Multiply neither the number of illustrations nor a single best-case modeling time by the desired roster. A different body plan or evolved silhouette may require substantial new animation and review.

Ownership and review coverage

Assign a named creative approver, engineering lead, art/pipeline owner and test coordinator. Before kickoff, document access to source files, permitted use of supplied assets, delivery rights and who supplies missing specialist work. Keep these as commercial agreement items to resolve with the partner, not assumptions hidden inside the schedule.

The handoff and the next decision

Accept a useful body of evidence, not just a visually impressive demonstration.

Required item	Acceptance evidence
Playable Windows build	Journey and systems-test modes; both input schemes; no open progression blocker.
Unity source	Exact editor/packages, .meta files, rule data, scenes, test fixtures and reproducible setup.
Blender source package	Editable pilot source, textures, export preset, clip manifest and integrated prefab references.
Design record	Implemented rules, deviations from this blueprint and rationale; current roster/scope.
Validation report	Functional matrix, raw player counts, observed confusion, performance captures and known issues.
Next-stage proposal	Separately estimated slice scope, risks, art cost per representative asset and staffing assumptions.

Go / revise / stop

Go: the functional checks pass, the primary comprehension targets are met, the pipeline reproduces reliably and the slice can be estimated. **Revise:** evidence points to a bounded issue with a clear fix and retest plan. **Stop or re-scope:** the core interaction remains weak or the required quality cannot be delivered within the available resources. Record who made the decision and what evidence supports it.

The first five working days

Day 1: confirm the project/repository, ownership of supplied material, scope and reference machine. **Day 2:** pin versions; make Boot, AssetLab and a standalone build. **Day 3:** round-trip one simple animated mesh from Blender; record settings. **Day 4:** create the four species records and a rules-only damage/target test. **Day 5:** show the first arena skeleton, review risks and replace the illustrative estimate with the team's plan.

The desired final proof: a player forms a Contract, makes a meaningful Conduit choice, earns a useful bond reward, and wants to set out with that companion again.

This PDF prepares that work. Building the Unity project and producing the game-ready Blender assets are the next implementation stage.

Canon and decisions to preserve

Maintain the identity of Astral Wilds while narrowing what is built first.

Term	Meaning in this project
Avelora / Astral Wells	The setting; Wells are concentrations of leylines that shape habitats and magic.
Magi / Conduit	The player's magical role; the wand-like focus through which support and Contracts are performed.
Astral / Contract / Bond	Creature; magical recruitment relationship; persistent companionship progression.
Orders / Convergence	Ten schools of magical philosophy; the broader competitive and narrative journey.
Eleventh point	The opening mystery: an impossible extra point in the first sigil. Its combat type is not defined here.
Resonance	A future temporary high-bond transformation; excluded from the proof.

Ten recognized types: Ember, Tide, Verdant, Terra, Gale, Volt, Frost, Radiant, Umbral and Arcane. The prototype activates only four. Starter base types are Cindrel/Ember, Mossling/Verdant and Ripplefin/Tide; dual-type art labels require a life-stage alignment pass before production.

Six Sovereigns: Solaryn (Radiant/Ember), Nyxara (Umbral/Arcane), Thaloros (Tide/Arcane), Terrakon (Terra/Verdant), Vaelstrix (Gale/Volt) and Velcryos (Frost/Umbral). They remain mythological references, not promised prototype enemies.

Outstanding decisions and owners

Creative lead: final Mossling proportions, evolved-form name and eleventh-point story meaning. **Design lead:** tune damage, Focus and Bond from observations; confirm whether the all-active-defeated rule is acceptable. **Technical lead:** exact Unity/Blender versions, compatible packages and actual hardware. **Producer:** team availability, quote, source-delivery terms and review cadence.

Reference-art handling

The supplied realistic single-creature studies are generally unnamed; do not silently relabel them as named starters or Sovereigns. Some original sheets still use the old project title. Use Astral Wilds in new production metadata and approved sheets. Maintain an asset register with source, creator/provenance notes, permitted use and production status.

Source basis and technical references

Primary documentation supports the workflow; design choices remain project proposals.

Project sources

P1. Astral_World_Game_Concept.docx, supplied by the creator. Foundation for Avelora, Magi, Astrals, Contracts, Conduits, types, Orders, Bond, evolution and Sovereigns.

P2. Astral_World_Final_Presentation.pptx and supplied Astral concept images. Visual direction and original presentation content.

P3. Astral_Wilds_Developer_Concept.docx and Astral_Wilds_Attention_Edition.pptx, produced in this task. Basis for the four-active/six-carried prototype, shared Focus, Warden/Mender, scoped counts and proposed player-test thresholds.

P4. User instruction: use Blender and Unity. This fixes the tool direction; the exact versions and production configuration remain discovery decisions.

Primary technical sources, checked 15 September 2026

[S1] Unity 6 release support

Release/support policy and the proposed LTS baseline. The installed newer editor is not automatically a reason to migrate an existing project.

[S2] Blender manual: FBX

Interchange, animation baking and material limitations. The cited manual is version 5.0; export behavior must be checked in the pinned Blender build.

[S3] Unity manual: Importing a model

FBX workflow, native authoring-tool dependency, and import settings.

[S4] Unity manual: ScriptableObject

Authored shared data and the distinction from deployed-game save persistence.

Links are clickable in the PDF. Unity manual references here use the documented Unity 6.0 pages for foundational concepts, not a claim that every package or Inspector control is identical in every Unity 6 release.

Implementation references and status

Keep the evidence boundary visible when this document is shared.

[S5] Unity manual: Introduction to URP

Rendering-pipeline context; the choice of URP and project quality settings is proposed here.

[S6] Unity manual: Generic animations

Non-humanoid animation setup and explicit root-node configuration.

[S7] Unity Input System: Using Actions workflow

Action-based input workflow. Pin a compatible released package rather than using a development branch as a dependency.

[S8] Unity manual: Profiling your application

Profiling a player build and separating diagnostic capture from final performance validation.

What was inspected for this document

The task's concept review, revised concept text and detailed pitch notes were reviewed. Local installation inspection found Unity editor 6000.6.0f1. Blender 5.2.1 LTS had already been used for the opening-film encoding. These checks establish available tools and source context; they do not establish that the game systems or game-ready assets have been implemented.

Version	Change / status
1.0 / 15 Sep 2026	First detailed Blender + Unity proof-of-concept blueprint. Introduces explicit testable battle values, readiness rules, persistence policy, pipeline gates and delivery criteria.
Next revision	Replace illustrative effort and toolchain choices with the development team's discovery decisions; add measured results and approved prototype deviations.

Readiness summary

Available: world and game concept, developer pitch, Astral visual references and a concept-art opening film.

Specified here: bounded prototype, example rules, Blender/Unity workflow, testing and handoff.

Next evidence required: a runnable Unity build, repeatable creature integration, observed player results and a costed slice proposal.

Begin with the first playable bond. Let its results determine how far Avelora grows.